Project Design Phase

Proposed Solution Template

Date	22 July 2026
Team ID	
Project Name	Crop Production Analysis for Indian Agriculture
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers and policymakers in India lack an easy, data-driven way to analyze historical crop production trends and predict future yield across states and crops, which leads to suboptimal crop planning, low income and avoidable crop losses.
2.	Idea / Solution description	A web-based Crop Production Analysis platform that ingests historical agricultural data (state/district-wise crop area, production, yield and rainfall), applies data analytics and machine-learning models to identify trends, and predicts future crop production, presenting the insights through an easy-to-use dashboard for farmers and departments.
3.	Novelty / Uniqueness	Combines multi-source data (historical yield, rainfall, soil and season) with ML-based production prediction and a simple, regional-language dashboard designed specifically for India's diverse agro-climatic zones, rather than a generic one-size-fits-all analytics tool.
4.	Social Impact / Customer Satisfaction	Better crop planning reduces farmer losses and improves income, supports food-security planning at the state/national level, and helps government departments design more effective subsidies and schemes.
5.	Business Model (Revenue Model)	Freemium access for individual farmers, subscription/licensing for state agriculture departments and agri-businesses, and paid data-insight partnerships with seed/fertilizer companies and agri-tech startups.
6.	Scalability of the Solution	Modular architecture allows new states, crops and data sources to be added easily; the platform can scale from a state-level pilot to a pan-India system and can be extended to other countries with similar agro-climatic datasets.